PEGASUS

Status Report 2

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Project Overview

- The main purpose of this device is to take a fuel, increase that fuel's efficiency, and significantly reduce emissions
- Secondary purpose is to be able to run on a wide variety of fuels, both petroleum and non-petroleum products
- Compact and suitable for many different sized applications

Review of status at Midterm

- In process of assembling reformer
  - Only main body completed
  - Majority of parts on hand awaiting assembly
- Some specialty parts not yet purchased
  - Part size needs adjustment, additional attachments needed for modification
- Copper fittings not yet soldered

Completed Since Midterm

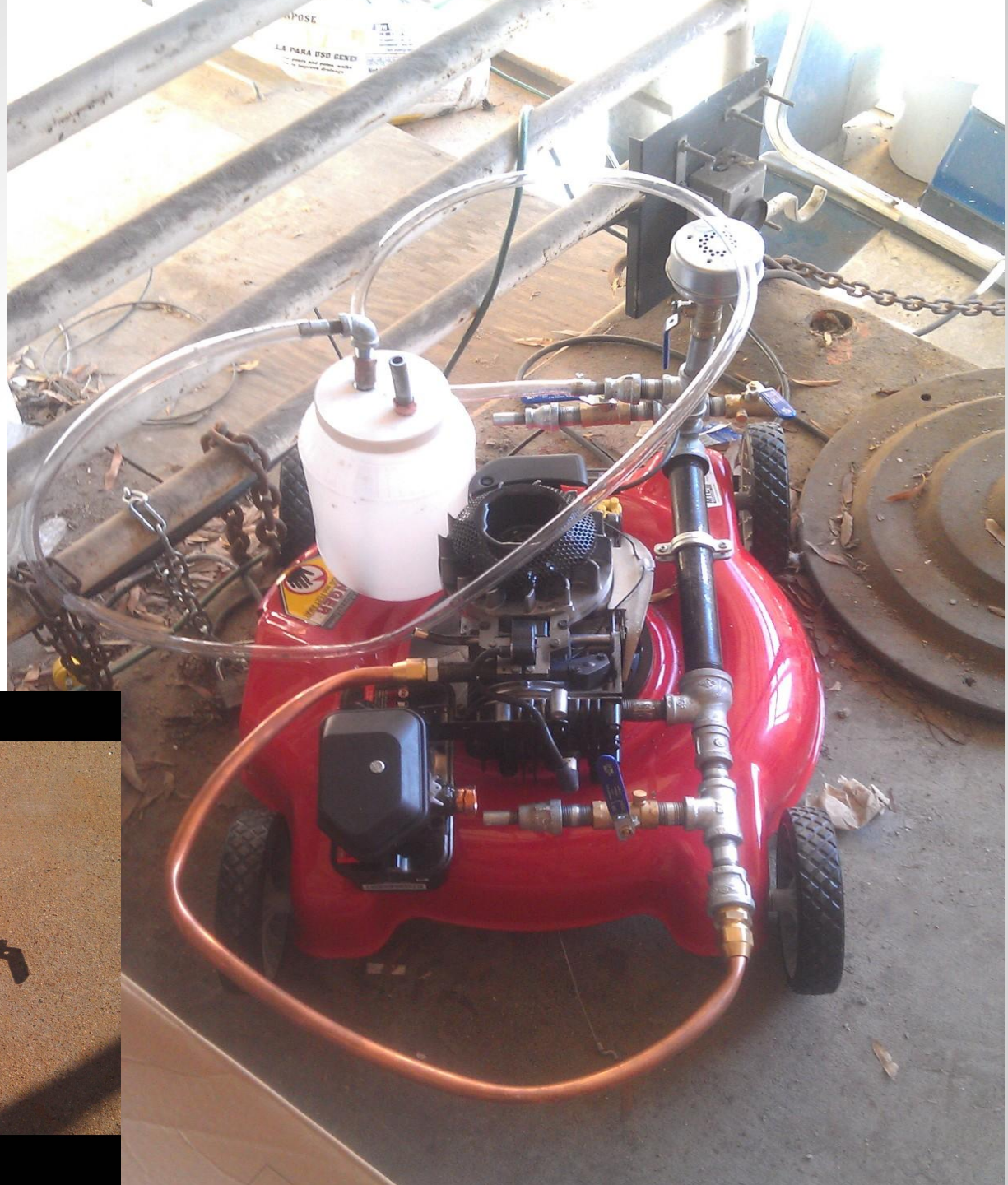
- Final purchasing of materials
  - Underestimated quantity of several parts
- Reformer fully assembled
  - Adjustments made to accept multi fuel sources
- Mounting of reformer to engine
  - Required removal of lawnmower handle; will remount if necessary
- Compression testing of system

Modifications to Design

- Addition of gasoline intake
  - Original fuel tank and carburetor utilized
- Connection to intake
  - Bonded using epoxy (not threaded as expected)
- Removal of secondary air intake

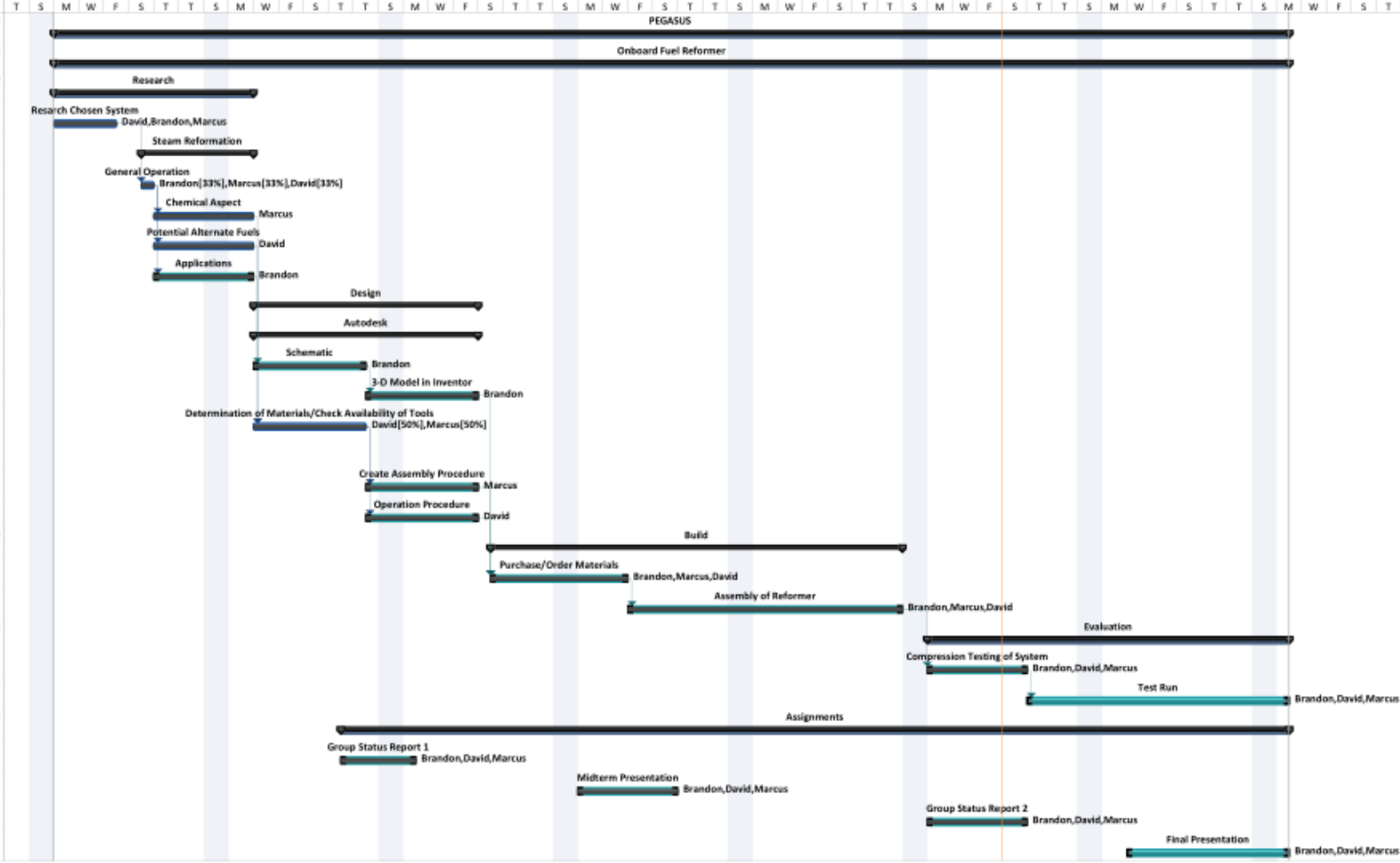
On-board Fuel Reformer

DANGER

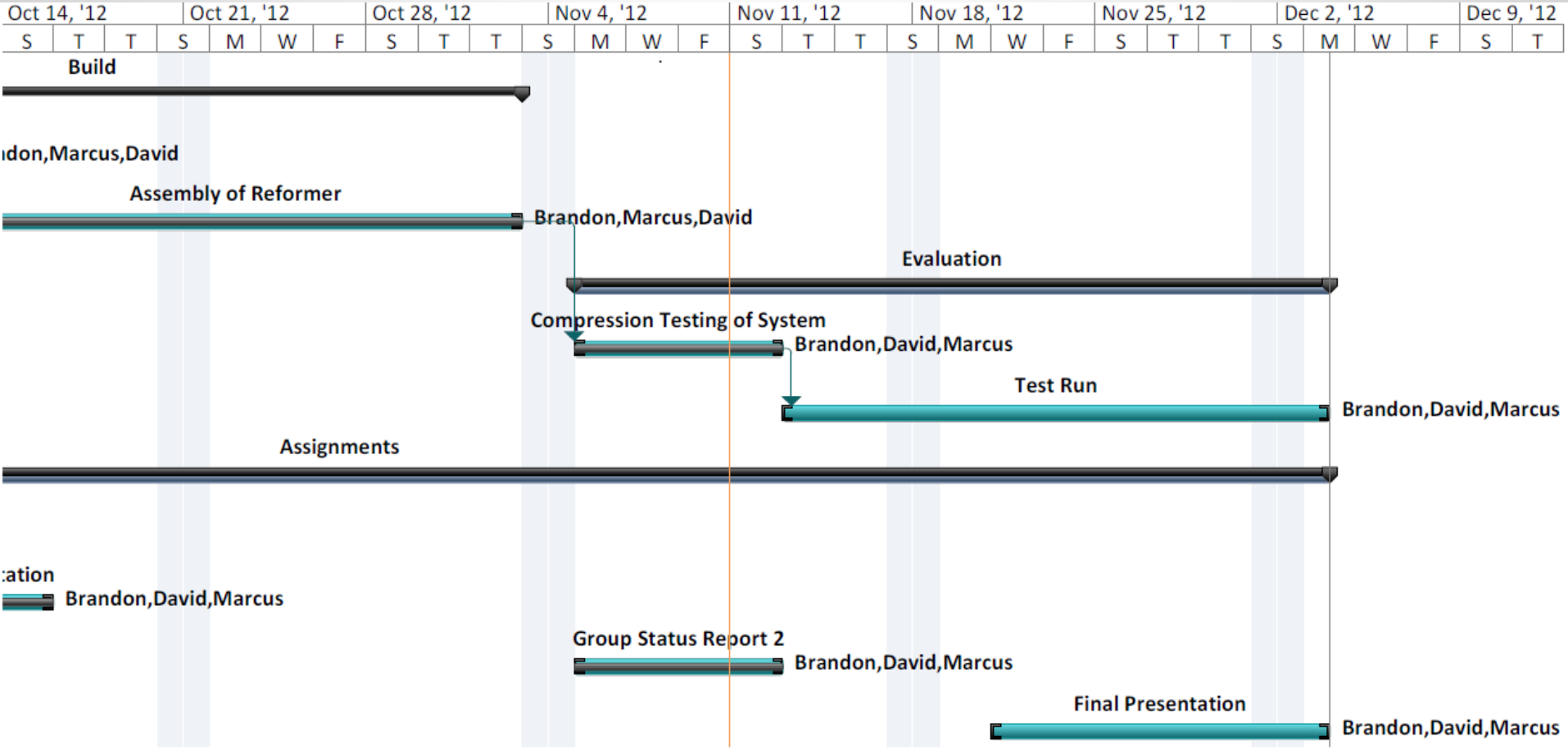


Gantt Chart Overview

ID	Task Name	Duration	Start
1	**PEGASUS**	**72 days**	**Mon 8/27/12**
2	**Onboard Fuel Reformer**	**72 days**	**Mon 8/27/12**
3	**Research**	**12 days**	**Mon 8/27/12**
4	Resarch Chosen System	5 days	Mon 8/27/12
5	**Steam Reformation**	**7 days**	**Mon 9/3/12**
6	General Operation	1 day	Mon 9/3/12
7	Chemical Aspect	6 days	Tue 9/4/12
8	Potential Alternate Fuels	6 days	Tue 9/4/12
9	Applications	6 days	Tue 9/4/12
10	**Design**	**14 days**	**Wed 9/12/12**
11	**Autodesk**	**14 days**	**Wed 9/12/12**
12	Schematic	7 days	Wed 9/12/12
13	3-D Model in Inventor	7 days	Fri 9/21/12
14	Determination of Materials/Check Availability of Tools	7 days	Wed 9/12/12
15	Create Assembly Procedure	7 days	Fri 9/21/12
16	Operation Procedure	7 days	Fri 9/21/12
17	**Build**	**25 days**	**Mon 10/1/12**
18	Purchase/Order Materials	9 days	Mon 10/1/12
19	Assembly of Reformer	16 days	Fri 10/12/12
20	**Evaluation**	**21 days**	**Mon 11/5/12**
21	Compression Testing of System	6 days	Mon 11/5/12
22	Test Run	15 days	Tue 11/13/12
23	**Assignments**	**55 days**	**Wed 9/19/12**
24	Group Status Report 1	4 days	Wed 9/19/12
25	Midterm Presentation	6 days	Mon 10/8/12
26	Group Status Report 2	6 days	Mon 11/5/12
27	Final Presentation	9 days	Wed 11/21/12



Gantt Chart


Oct 14, '12
Oct 21, '12
Oct 28, '12
Nov 4, '12
Nov 11, '12
Nov 18, '12
Nov 25, '12
Dec 2, '12
Dec 9, '12
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Assembly of Reformer
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Evaluation
Compression Testing of System
Brandon,David,Marcus
Test Run
Brandon,David,Marcus
Assignments
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Brandon,David,Marcus
Group Status Report 2
Brandon,David,Marcus
Final Presentation
Brandon,David,Marcus

Future Tasks through Final

- Testing of the system
  - First run using gasoline intake
  - Reformer operation using methanol
- Modifications as needed
- Completion of documents
  - Safety procedure
  - Operation procedure

Questions?